

Detim Zhao

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Education

Bachelor of Science, Software Engineering

May 2026

Arizona State University, Polytechnic – GPA: 3.90 / 4.00

Mesa, AZ

- Barrett (Honors), Dean's List (Fall 2022, Spring 2023, Fall 2023, Spring 2024)

Technical Skills

Languages – Java, Python, C, C++, PostgreSQL, MySQL

Technologies – pandas, scikit-learn, matplotlib, seaborn, NumPy, PyTorch

Tools – Git, GitHub, LaTeX, Unreal Engine 5 (UE5), Photoshop, Illustrator, Excel, Gazebo, Robot Operating System, Mavsdk, PX4 Autopilot

Experience

Extended Reality (XR) Creative Developer

July 2024 – Present

Meteor Studio at ASU

Mesa, AZ

- Developing a research project in UE5 to model a sensor-equipped drone following a runner on a track by utilizing UE5's sequencer and blueprints

Undergraduate Researcher

June 2023 – Aug 2023

Texas A&M-Corpus Christi

Corpus Christi, TX

- Partnered to design and develop a Python program to operate multiple PX4 drones in a gazebo simulation through terminal controls, achieving successful control of up to 8 to 10 drones. (Python)
- Applied preprocessing to two datasets, then trained and tested five ML models (KNN, SVM, Random Forest, Naive Bayes, Ensemble) for gesture recognition, enhancing classification accuracy by 10% (Python)
- Co-authored a research paper for publication (DOI: [10.1145/3565287.3617621](https://doi.org/10.1145/3565287.3617621)) that documents findings and work, leading to publication in ACM and conference presentations (LaTeX)

Projects

Gesture Recognition of Hand Gestures with ML

- Streamlined data preprocessing, including data cleaning and handling missing values on small and large incomplete datasets, enhancing nearly 15% of data for ML usage.
- Trained and evaluated five different ML models on small and large datasets, enhancing model accuracy by about 10% through hyperparameter tuning. (Python, scikit-learn).
- Compared algorithms of each model from small and large datasets using standardized benchmarks, identifying the best-performing models and organizing information for the published paper.

Control Drone (PX4) Swarm in Gazebo Simulation

- Designed a script to automate setup for PX4 drones and Gazebo, reducing manual inputs and setup time by 30% and optimizing future testing. (Python)
- Implemented terminal input commands to control individual drones or groups of drones as a flexible control system, increasing operational efficiency and control accuracy.
- Developed a logging mechanism to track drone movements, enhancing debugging and analysis by 25%.

Console-Based Chat Application

- Developed console-based directions and options for users to register, log in/out, update account info, create a chat room, and etc., streamlining user interactions.
- Implemented commands when the user is in a chat room to display active users, chat history, and more, improving overall user experience.
- Designed and Organized local host pSQL database to store information about the user and messages, ensuring efficient data storage and queries. (Java, PostgreSQL)

Maze Generator and Solver

- Developed a GUI-based maze generator and solver where the user specifies the dimensions of the maze, improving user experience and usability.
- Implemented a recursive backtracking algorithm to solve the maze and draw the solution path, successfully solving and visually representing maze solutions with the GUI. (Java, JavaFX)